Bags, Vectors & Transformers

Synthetic Data and Augmentation with LLMs · Bonus, Part 19

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What Synthetic Data Is For

	The LLM writes	Typical goal
Augmentation	rewordings of your coded examples	more variety from the same labels
Synthetic training data	new examples for each category	labels where you have few, e.g. a rare category
Test cases	targeted variations	probe what a classifier gets wrong
Shareable stand-ins	texts that resemble confidential ones	teach or share without the originals

Synthetic data trains, real data tests. Every score in this module is on real, human-coded titles.

Not synthetic respondents. LLMs standing in for survey respondents are a separate and contested idea (Argyle et al., 2023; Bisbee et al., 2024).

Three Ways to Get More Training Data

Paraphrase

Reword each coded title five times. The label stays the same.

40 coded titles become 240

From definitions

Give the LLM the codebook definition of an area and ask for new titles.

no real examples at all

With examples

The definition plus three real titles to show the style, and an instruction not to copy them.

few-shot generation

```
Write 10 realistic, varied titles of US Congressional bills
in the policy area 'Taxation':  taxes of any kind, ...
For the style, here are some real titles.  Don't copy them:
```

Qwen2.5 1.5B with sampling (temperature 0.9): for training data we **want** variety.

The Experiment: Only 40 Coded Titles

Training set	texts
40 real titles (10 per area)	40
840 real titles (the ceiling)	840
40 real + paraphrases	240
synthetic, from definitions	200
synthetic, with examples	200
40 real + synthetic	240

TF-IDF classifier (Day 1), the same 100 real test titles as Day 4.

What studies find

Synthetic data helps more for objective tasks than for subjective ones, and generating with real examples beats generating without (Li et al., 2023).

Across ten social science tasks, human-coded data did as well or better; synthetic data helped mostly for **rare classes** (Møller et al., 2024).

Is the Synthetic Data Any Good? Four Checks

Diversity

Average similarity between pairs of texts. Lower means more varied.

Realism

Can a classifier tell synthetic from real? AUC 0.5: no; close to 1: easily.

Label quality

Does a classifier trained on real titles put each synthetic title in its intended category?

Leakage

Are synthetic titles near-copies of test titles, or of real bills the LLM has seen?

Typical pattern: more alike than real texts, easy to spot, and clearly in their category. Clean labels, but few of the **borderline cases** where classifiers make their mistakes.

Risks

- **The model's view becomes your data**, stereotypes included (Day 4, ethics)
- **Too clean, too similar**: prototypes teach the easy cases
- **Contamination**: "new" texts can be memorized real ones, including your test set
- **Model collapse**: models trained on their own output, generation after generation, lose the rare cases (Shumailov et al., 2024)
- **Privacy is not automatic**: texts generated from confidential data can repeat its details

Temperature is a trade-off. Low: repetitive but on topic. High: varied but more off-topic or garbled. Filtering with a classifier afterwards makes its blind spots the data's blind spots.

Good Practice

Do

- start from your own coded examples
- add synthetic data to real data, don't replace it
- use it where it helps most: rare categories, prototypes, test cases
- choose how much on development data

Report

- which texts are synthetic, and how many
- model, prompts, temperature, filtering
- the quality checks and their results
- scores on real, human-coded data only

If you can code more real data, that is usually the better investment.

In the Notebook

- A hard starting point: 10 coded bill titles per policy area
- Augmenting them with LLM paraphrases
- Synthetic titles from definitions, and with real examples
- Six training sets, one real test set: did it help?
- Diversity, realism, label quality and leakage checks
- Temperature, rare categories and memorized titles

`bonus/notebooks/19_synthetic_data_and_augmentation_exercises.ipynb`
5 exercises · GPU